

Wireless VU METER

VISWESWARA RAO KALLA

VU meter is a standard device that indicates signal level in audio equipment. Unlike other devices, this meter doesn't require physical connection between the audio source and the circuit.

Circuit and working

Circuit diagram of the wireless VU meter is shown in Fig. 1. It is built around an ATmega32A IC (IC1), a bargraph LED (BAR1), a 1N4148 diode (D1), transistors BC547 (T1) and BD139 (T2), and a few other components.

Transistor T2, diode D1, zener diode ZD1 and passive components form the linear voltage regulator. A 7V-12V, 200mA power source is connected to connector CON1. Diode

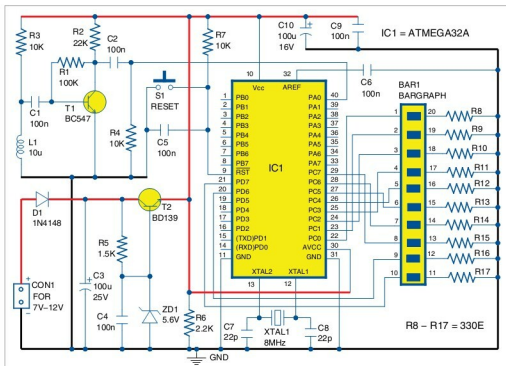


Fig. 1: Circuit diagram of wireless VU meter

PARTS LIST

Semiconductors:	
IC1	- ATmega32A microcontroller
BAR1	- 20-pin bargraph LED
T1	- BC547 npn transistor
T2	- BD139 npn transistor
D1	- 1N4148 diode
ZD1	- 5.6V zener diode
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 100-kilo-ohm
R2	- 22-kilo-ohm
R3, R4, R7	- 10-kilo-ohm
R5	- 1.5-kilo-ohm
R6	- 2.2-kilo-ohm
R8-R17	- 330-ohm
Capacitors:	
C1, C2, C4-C6, C9	- 100nF ceramic disk
C3	- 100µF, 25V electrolytic
C7, C8	- 22pF ceramic disk
C10	- 100µF, 16V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector
XTAL1	- 8MHz crystal
S1	- Push-to-on switch
L1	- 10µH ferrite-core inductor
	- 20-pin IC socket for BAR1
	- 7V-12V, 200mA DC power supply

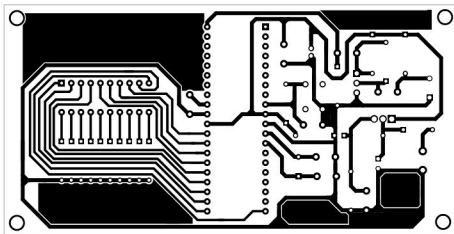


Fig. 2: Actual-size PCB layout of wireless VU meter

D1 is used to protect the regulator from reverse voltage connection and capacitor C3 reduces the noise in the power supply lines. Resistor R5 sets the zener diode (ZD1) current. The output voltage at the emitter of T2 is around 4.9V.

A preamplifier or small-signal amplifier is built around transistor T1, where resistor R2 sets the collector current, R1 is the feedback resistor to set the proper gain, and capacitors C1 and C2 are the input and output coupling capacitors, respectively.